

CAPE NELSON, Australia

WMO: 948260

Lat: 38.4306S

Lon: 141.5436E

Elev: 153

StdP: 14.61

Time Zone: 10.00 (AUS)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.9	44.8	35.2	30.2	50.8	36.7	32.2	50.7	36.4	52.8	32.7	53.0	8.8	0	0.651

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.8	84.8	65.1	79.4	63.0	74.5	62.5	68.1	77.3	66.3	73.6	64.8	70.9	14.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	96.8	69.5	64.2	90.4	67.7	62.8	85.9	66.7	32.5	77.2	31.1	73.6	30.0	71.2	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.2	27.7	24.9	DB	38.8	99.4	1.5	4.5	37.7	102.6	36.9	105.3	36.1	107.8	35.0	111.0	(4)
(5)			WB	37.2	72.7	1.7	4.0	36.0	75.6	35.0	77.9	34.1	80.1	32.8	83.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.8	63.7	64.1	62.5	59.2	56.2	52.9	52.0	52.6	54.4	56.6	58.7	60.9	(6)	
	DBStd	5.94	5.03	4.86	4.89	4.17	3.48	2.77	2.62	3.06	3.92	5.03	4.99	4.91	(7)	
	HDD50	29	0	0	0	0	1	6	11	8	2	1	0	0	(8)	
	HDD65	2809	84	64	110	183	275	362	403	384	318	271	205	150	(9)	
	CDD50	2873	423	396	387	277	193	94	73	89	136	205	262	338	(10)	
	CDD65	177	42	40	32	10	2	0	0	0	2	10	16	23	(11)	
	CDH74	1197	286	220	215	58	4	0	0	0	8	98	128	180	(12)	
	CDH80	440	111	88	84	15	0	0	0	0	0	28	42	72	(13)	
(14) Wind	WSAvg	13.7	13.2	13.3	12.7	12.2	13.3	13.4	14.7	14.6	14.3	14.5	13.9	13.7	(14)	
(15) Precipitation	PrecAvg	30.5	1.4	1.1	1.7	2.3	3.0	3.9	4.4	3.9	3.2	2.4	1.9	1.7	(15)	
	PrecMax	38.0	4.3	4.3	6.7	6.3	7.6	7.4	8.9	8.0	5.6	4.8	4.8	4.1	(16)	
	PrecMin	20.1	0.2	0.1	0.2	0.3	0.8	0.9	2.1	0.7	1.1	0.2	0.6	0.2	(17)	
	PrecStd	4.5	1.1	0.9	1.1	1.3	1.6	1.4	1.5	1.5	1.2	1.3	0.9	1.0	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.9	90.9	89.6	80.9	71.3	63.5	61.8	68.0	74.5	83.7	85.9	89.2	(19)	
		MCWB	66.3	67.2	65.3	61.6	58.3	55.7	53.9	56.0	59.7	60.9	63.7	67.6	(20)	
	2%	DB	81.5	80.4	79.6	73.2	66.7	60.1	58.9	62.7	68.7	75.2	77.0	78.5	(21)	
		MCWB	65.7	64.3	62.7	59.7	57.0	54.9	53.2	54.0	56.7	58.7	61.8	63.0	(22)	
	5%	DB	74.7	73.9	72.5	67.9	63.4	58.5	57.3	59.6	63.9	68.8	70.4	71.4	(23)	
		MCWB	65.0	64.7	62.6	58.7	56.5	54.1	52.5	53.2	55.7	57.5	61.2	63.1	(24)	
	10%	DB	69.8	70.0	67.9	64.5	61.0	57.3	55.9	57.3	60.3	63.4	65.5	66.7	(25)	
		MCWB	63.9	64.0	62.0	58.7	56.3	53.3	51.7	52.4	54.1	56.5	60.0	61.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.0	70.7	67.8	64.4	60.9	58.2	57.2	58.3	62.1	63.0	67.1	70.3	(27)	
		MCDB	76.3	80.0	78.5	73.3	65.1	60.3	59.3	62.7	68.0	76.5	77.1	78.6	(28)	
	2%	WB	68.1	67.7	65.5	62.4	59.4	56.4	54.8	55.8	58.8	60.5	64.0	65.7	(29)	
		MCDB	76.2	76.4	73.4	67.1	62.9	59.0	57.0	59.7	65.3	71.1	71.7	73.4	(30)	
	5%	WB	66.0	65.9	64.0	61.0	58.2	55.0	53.5	54.5	56.5	58.7	62.1	63.5	(31)	
		MCDB	72.5	71.4	69.9	65.2	61.6	57.6	56.0	58.1	62.1	65.9	68.7	69.8	(32)	
	10%	WB	64.1	64.5	62.6	59.6	57.1	53.9	52.5	53.2	55.0	57.0	60.2	61.7	(33)	
		MCDB	69.4	69.0	67.1	63.8	60.4	56.7	55.3	56.7	59.6	62.5	65.4	66.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	8.8	9.0	9.0	7.8	7.6	7.6	8.5	9.8	10.5	9.5	9.7	(35)	
		MCDBR	21.5	18.7	19.9	17.2	12.4	9.6	9.7	12.9	17.5	22.9	21.5	21.4	(36)	
	5% WB	MCWBR	9.1	8.0	8.1	8.7	7.6	7.2	6.8	8.0	9.6	10.4	9.6	9.7	(37)	
		MCWBR	16.3	14.3	14.7	12.4	9.7	8.2	8.3	10.9	14.7	18.2	17.0	16.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.341	0.340	0.328	0.321	0.309	0.309	0.304	0.317	0.342	0.340	0.338	0.339	(40)	
		taud	2.554	2.565	2.601	2.603	2.617	2.595	2.612	2.573	2.486	2.500	2.533	2.549	(41)	
	Ebn at Noon		314	307	298	280	265	254	264	278	288	304	313	317	(42)	
		Edh at Noon	34	32	29	25	21	20	21	25	31	34	34	34	(43)	
(44) All-Sky Solar Radiation	RadAvg		2240	1902	1496	1048	705	556	611	885	1287	1719	2008	2227	(44)	
	RadStd		115	107	76	69	38	35	34	46	85	114	123	146	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.61	N/A	N/A	+2.30	+0.97	+1.37	N/A	-212	+272	+61
Regional (19 neighbors)		+0.76	N/A	N/A	+1.73	+0.76	+0.90	N/A	-167	+227	+63

Nomenclature: See separate page